

Year 10 General



Baldivis
Secondary College

Mini Test 1 Index Laws and Algebraic Fractions

Name: Answers

Score: 14 / 28
Total Time: 35 minutes

Show working and answers on this sheet. Show working in sufficient detail to support your answers. Incorrect answers without supporting reasoning may be allocated zero marks. Calculators may be used. 1 side A4 notes (handwritten) allowed.

- 1) Simplify the following expressions (leave your answers in index form)

9
(10 marks)

a) $15m^7 \div 3m^2 = 5m^5$ ✓ 1mk

b) $(2x^6y)^4 = 16x^{24}y^4$ ✓ 2mks

c) $2a^3 \times 3a^9 = 6a^{12}$ ✓ 1mk

d) $2j^3k^2 \times 4j^{-4}k^{-3} = 8j^{-1}k^{-1}$ ✓ 2mks

e) $\frac{4x^2y \times (3xy^4)^2}{2x^4y^3 \times 3y^8} = \frac{4x^2y \times 9x^2y^8}{6x^4y^{11}} = \frac{36x^4y^9}{6x^4y^{11}} = 6y^{-2}$ ✓
3 mks

- 2) Simplify the following expression

14
(10 Marks)

a) $\frac{5x}{3} + \frac{8x}{3} = \frac{13x}{3}$ ✓ 1mk

$$b) \frac{5x^3}{6} - \frac{3x^3}{6}$$

$$= \frac{2x^3}{6} = \frac{x^3}{3} \quad 2 \text{ mks}$$

$$c) \frac{2x}{5} - \frac{x}{3}$$

$$= \frac{6x}{15} - \frac{5x}{15} = \frac{x}{15} \quad 2 \text{ mks}$$

$$d) \frac{y}{2} - \frac{z}{3}$$

$$= \frac{3y}{6} - \frac{2z}{6} = \frac{3y-2z}{6} \quad 2 \text{ mks}$$

$$e) \frac{k}{8} \times \frac{4k}{3}$$

$$= \frac{4k^2}{24} = \frac{k^2}{6} \quad 2 \text{ mks}$$

$$f) \frac{x+3}{2} + \frac{2x+1}{5}$$

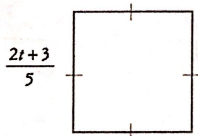
$$= \frac{5(x+3)}{10} + \frac{2(2x+1)}{10} = \frac{5x+15+4x+2}{10} = \frac{9x+17}{10} \quad 3 \text{ mks}$$

$$g) \frac{2k+1}{2} \div \frac{k+7}{8}$$

$$= \frac{2k+1}{2} \times \frac{8}{k+7} = \frac{8k+4}{k+7} \quad 2 \text{ mks.}$$

3) Find the perimeter and area of the following shape

5
(marks)



$$P = \frac{2t+3}{5} \times 4 = \frac{8t+12}{5} \quad 2 \text{ mks}$$

$$A = \frac{2t+3}{5} \times \frac{2t+3}{5}$$

$$= \frac{4t^2 + 12t + 9}{25} \quad 3 \text{ mks}$$

I am going to be fairly lenient here.
1 mk - some of it expanded properly
1 mk - all of it expanded properly